

Proposal quality and candidate evidence

The emailed leans now have explicit local candidate contracts.

The group still needs to review those choices. Independently CRS-bound point-instancer prototypes remain a model gap.

64 regressions and 47 distinguishing controls passed.
204,729 coordinate results per Python/C++ path.

Geographic scene frames/bounds remain Q9. Complete proposal conformance is not claimed.

Concrete contracts for group review

Subject	Concrete candidate answer
Q3: working frame	Nearest enclosing CRS fixes the adjustment chart. The candidate specifies origins, pivots and descendant resets.
Q6: axes and units	Z-up: (x,y,z). Y-up: (x,-z,y). Convert stage units before scale and orientation.
Q11: measurements	A proposed non-Xformable data-source schema selects asset, format, field and coordinate domain. Native formats retain values and times.
Q12: dependency	Profiles on defaultPrim carries a conservative hard capability claim for the whole scene, including unloaded content.

These are proposed details, including a new external-association schema. Passing tests does not establish group agreement.

A fixed working frame for project adjustments

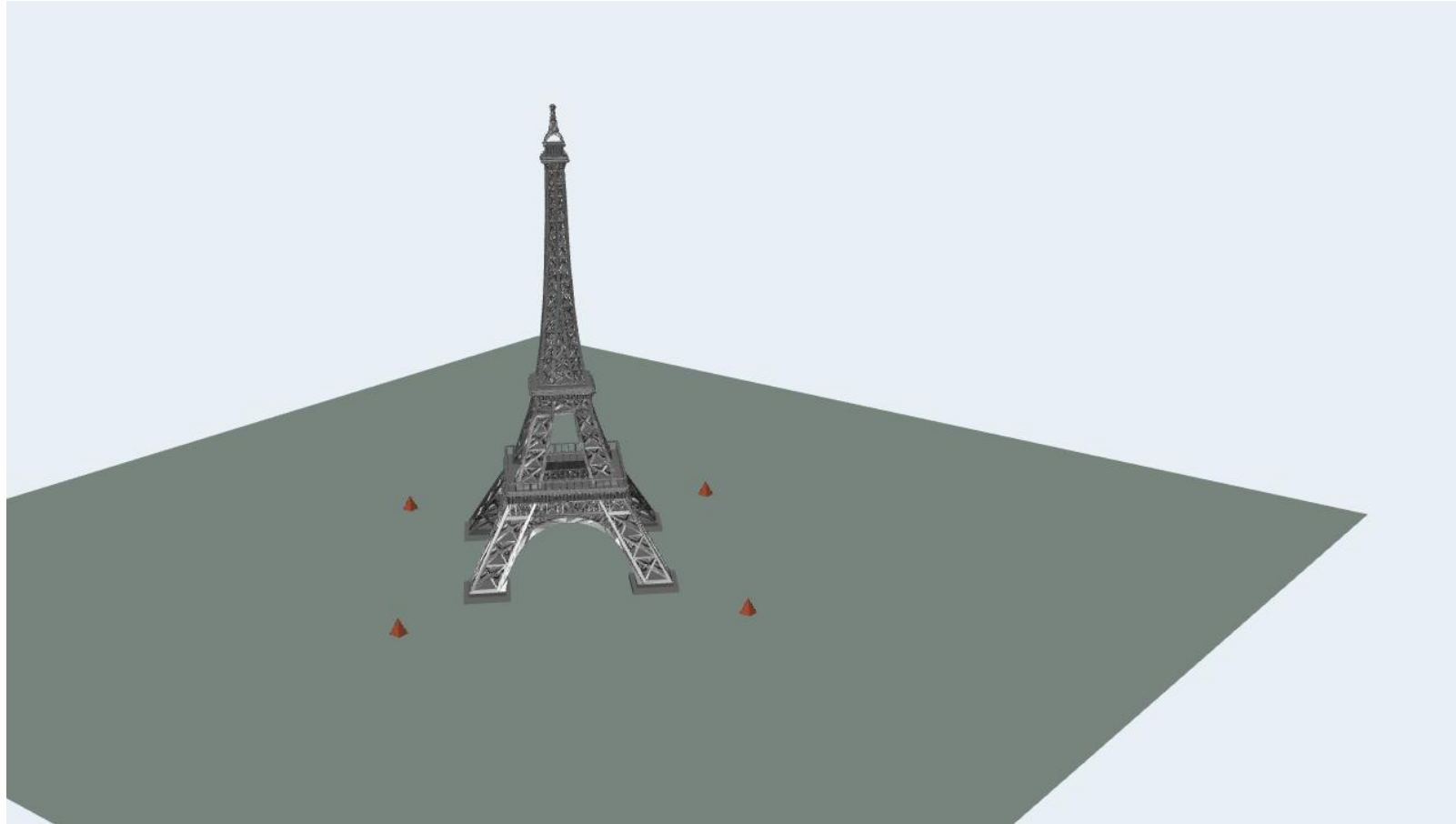
A geographic asset sits inside a projected site. Its model orientation turns local X northward.

Queried position	Easting (m)	Northing (m)	Height (m)
Anchor with +10 site adjustment	448262.0014	5411954.9099	35.0000
Child with +3 local X	448262.0292	5411957.9087	35.0000
Child with ordinary reset	448252.0292	5411957.9087	35.0000

The +10 follows site easting. The child offset follows the oriented model.
The reset removes the ordinary adjustment and retains CRS placement.

Reprojection to ECEF preserves the same adjusted physical result. Ordinary parent transforms remain excluded at a direct binding.

Eiffel geometry in the native Hydra consumer



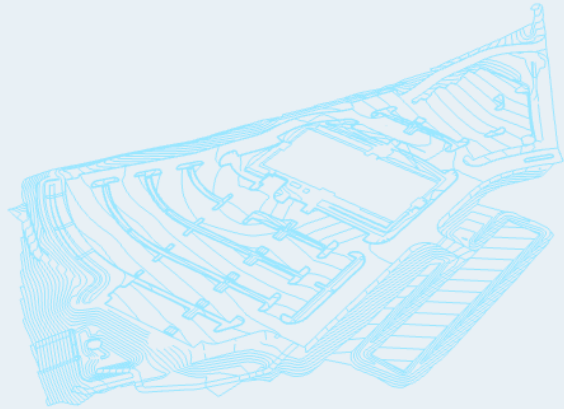
163,440 vertices

C++ pointwise results
Fresh USD export readers
Hydra points and
transforms
Native Storm color output

Illustrative plane and markers.
No surveyed Paris context.

The renderer consumes an ordinary resolved export. This run does not install a live geospatial scene-index filter.

Colorado survey geometry in a calibrated site



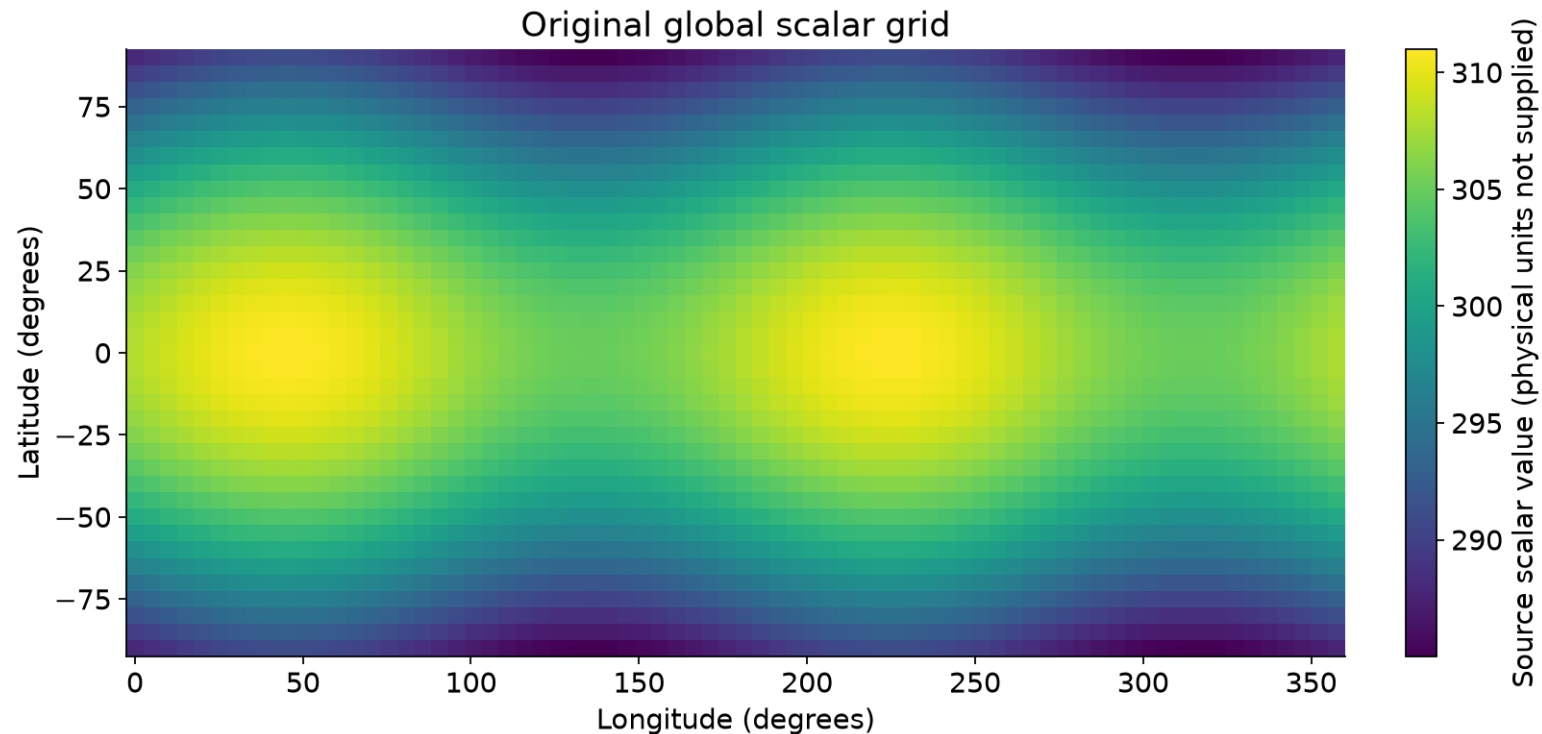
14,359 vertices

Original LandXML
breaklines
WKT-carried site
calibration
Declared US survey feet
Separate stage
conventions

No terrain surface
was fabricated.

Fresh readers verify the resolved geometry. Polygonal bounds cover returned vertices and straight faces, with no continuous-surface certificate.

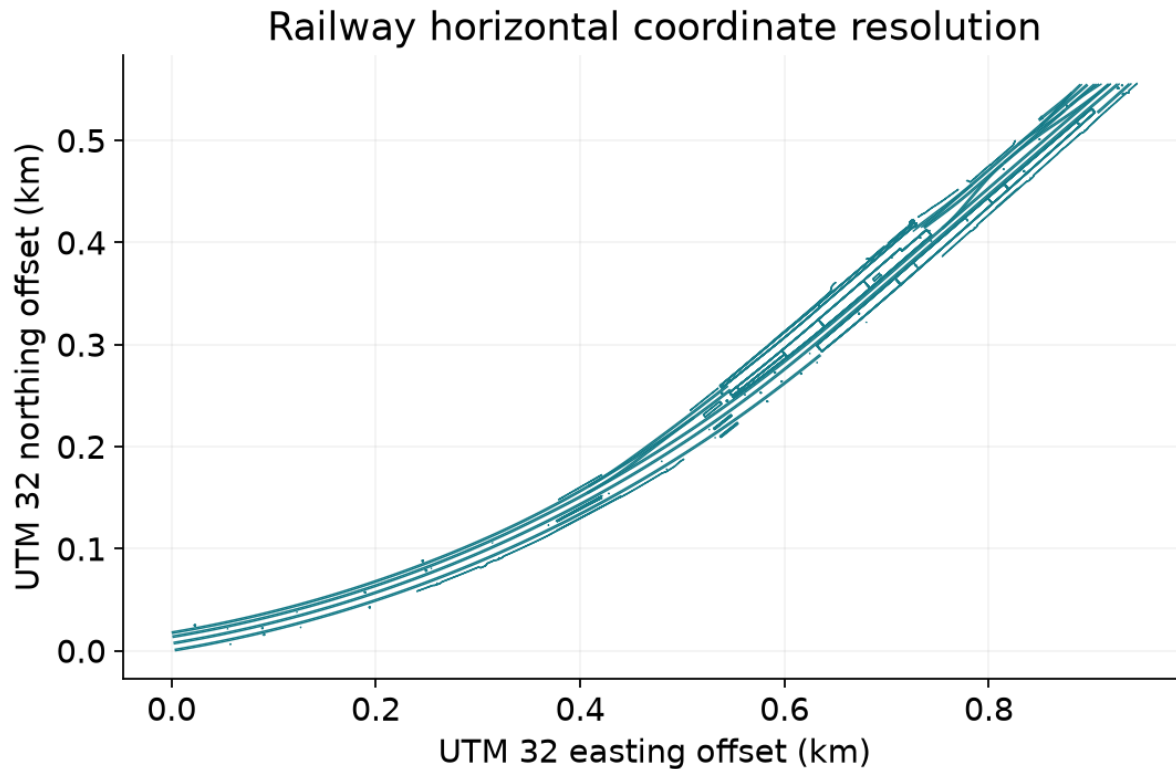
Georeferenced measurements remain usable as data



2,664 original values and horizontal coordinates remain paired.

The association invents no height, physical units or observation times. A 3D request fails without a height reference and coordinate.

Regional geometry retains its source coordinates



15,822 vertices

2,386 railway parts
Explicit horizontal source copy
UTM 32 output coordinates
Original data retained

Original third components
remain uninterpreted.

The larger-scale workflows keep useful 2D resolution without inventing height. Plot offsets serve display only.

Matching partner-coordinate queries

France: CC49 to Lambert-93 (metres)

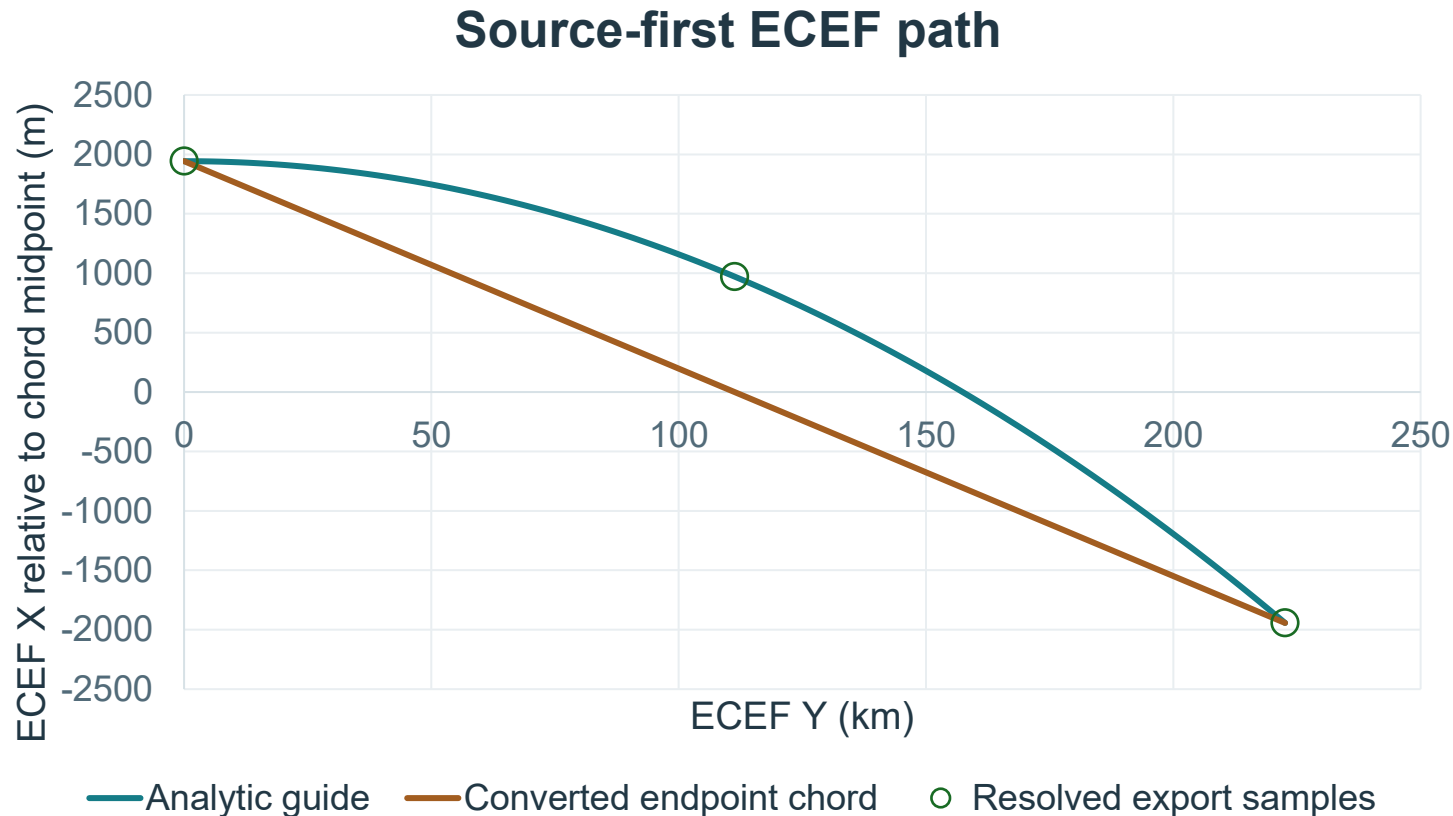
Component	Source	USD Python	USD C++	OV-hosted Python
Easting	1661099.0390	661101.1094	661101.1094	661101.1094
Northing	8180053.1170	6857834.2565	6857834.2565	6857834.2565
Height	37.2980	37.2980	37.2980	37.2980

Colorado: State Plane to calibrated site (US survey feet)

Component	Source	USD Python	USD C++	OV-hosted Python
Easting	3108002.1122	3108871.4501	3108871.4501	3108871.4501
Northing	1752265.0666	1206317.4694	1206317.4694	1206317.4694
Height	5482.9424	5484.6779	5484.6779	5484.6779

Both USD implementations agree. OV reuses Python placement. All paths share PROJ; these comparisons do not certify survey accuracy.

Source interpolation and sampled exports



971.421 m

Midpoint difference from interpolating only the converted endpoints.

Fresh geometry exports
Sample keys 0, 5, 10
timeCodesPerSecond = 48

Both readers verify each exported sample. No original-trajectory guarantee is claimed between samples. Chart axes use different distance scales.

Consumer evidence and remaining limits

Consumer	Evidence reaching its output
USD Python and C++	Independent placement arithmetic; full point maps, Cartesian frame estimates and polygonal bounds. Shared PROJ and external decoding.
Omniverse live geometry	177,846 vertices read back from runtime buffers/matrices. 27 edit, failure and recovery checks. Reuses Python placement.
Hydra / Storm	Two ordinary resolved exports reach points/transform data sources and the native renderer. No live CRS filter.

Eight exports pass fresh-reader checks. Authored source layers remain unchanged.

Remaining limits: bound point-instancer prototypes, geographic scene frames/bounds, continuous nonlinear certificates and physics integration. Candidate choices still need group review.